

Detection and Analysis of Cardiovascular Diseases using Machine Learning Techniques

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Abstract—Abstract—The detection and analysis of cardiovascular diseases is a critical study field within the medical field. Cardiovascular diseases are one of the leading causes of mortality globally, and timely identification is vital to forego adverse effects and improve outcomes for patients with these conditions. The incorporation of machine learning technologies to forecast onset of cardiovascular diseases and analyzing their progression has shown promising results. This paper demonstrates the machine learning models that can provide near-accurate predictions and help identify the right data set based on accuracy. The choice of the best-fitting algorithm has been made by a comparative study and with the inclusion of K-fold cross-validation. We infer that the Random Forest algorithm is the best algorithm for detection and K-Nearest Neighbor Algorithm for the analysis of prominent heart diseases like Stable Angina, ST-elevated myocardial infarction, and non-ST elevated myocardial infarction which will serve as a direction to further diagnosis. Overall, the potential for machine learning to improve early detection and treatment of Cardiovascular Diseases, which could ultimately save lives and reduce healthcare costs, has been recognized. **Index Terms**—Artificial intelligence, Machine learning, Supervised learning, Unsupervised learning, Accuracy

I. INTRODUCTION

Cardiovascular diseases (CVDs) are crucial concern for the public's health, and efficient treatment and management rely on prompt identification and accurate diagnoses. Traditional diagnostic methods for CVDs involve invasive procedures that can be high-priced and demanding on time. The use of machine learning (ML) algorithms in the detection and

analysis of CVDs has the potential to improve the accuracy and efficiency of diagnosis as well as enhance treatment and prevention strategies for these critical health conditions [1].

In the artificial intelligence space, machine learning is the study of developing algorithms that are capable of learning from data and employ it to predict or render decisions. It allows computers to automatically learn from historical data. Machine learning creates mathematical models using historical data or information and classifies the results or makes predictions.

By training ML algorithms on large patient data, healthcare professionals can potentially identify patterns and correlations that may be indicative of CVDs. The World Health Organization estimates that heart disease causes 17.9 million deaths worldwide each year [2]. Detection of cardiovascular diseases is crucial in order to start treatment with counseling and medications. The existing methods detect the presence of the disease, but there is no system as such to analyze the type of heart disease. Delayed detection can lead to complications in the heart health of the patients. Heart-related diseases can turn fatal if not diagnosed well in advance. This proposed method is well suited for use in the detection of major types of heart disease, namely, *stable angina* and *ST-elevation myocardial infarction* (STEMI).

This paper is split into five sections. Section I provides a brief summary to cardiovascular disease. Section II reviews the research work done in recent years on heart disease predictions. Section III discusses about the methodology followed for the prediction of *stable, angina* and *STEMI*. The obtained

results have been discussed in the Section IV. The final section concludes our current work and gives a prelude to future work.

II. RELATED WORKS

Recent years have witnessed extensive research on the detection of heart diseases by implementing various technologies.

Machine learning has shown to be beneficial in many important ways when used to identify different diseases.

Rahul Katarya and Polipireddy Srinivas [3] present a summary of some such automated systems. Every automated system needs to use feature selection and prediction. Selecting appropriate features can help improve the heart disease predictions. A couple of algorithms have been described, such as the hybrid grid search algorithm and the random search algorithm, that are helpful for choosing the features.

Mangesh Limbitote et al. [4] summarized a number of machine learning techniques and the attributes considered by various researchers. Every algorithm has produced a distinct outcome in a variety of circumstances. Further analysis shows that the predictive model for heart illness only achieves minimal accuracy; Therefore, in order to increase the precision of early heart disease prediction, sophisticated models must be developed.

Rohit Bharti et al. [5] have offered three techniques involving machine and deep learning models. These models were compared based on four measures including F1 score, specificity, sensitivity, precision and confusion matrix. The small size of the data set was found to be the reason why machine learning models outperformed deep learning models. The idea that machine learning models should be utilized over deep learning ones for small datasets is proved in the study. When data preprocessing is used, the KNeighbors classifier outperformed the ML technique for the 13 features that were included in the dataset. Additionally, the computing time was cut down, which is beneficial for model deployment. Likewise, it was noted that in the absence of normalization, the model is prone to over-fitting and achieves insufficient accuracy when applied to real-world data problems.

Many authors [6], [7] have used Naive Bayes and decision trees to predict heart diseases. While some of them find that Naive Bayes are most ideal for classification of their data [6], others find that the decision tree-based algorithms achieve better results for their data set [7]. Lutimath et al. [8] used Naive Bayes classification and the radial kernel support vector machine on UCI Cleveland dataset and observed that the Naive Bayes classifier provided the most accurate prediction. Researchers [9]–[11] have also used random forests for classification of heart diseases

III. METHODOLOGY

A. Data Collection

We have considered the following two different datasets from Kaggle:

- 1) 'Heart Disease Dataset' [12] and
- 2) 'Cardiovascular Diseases Dataset (Clean)' [13].

The first data-set contains clinical characteristics related to heart disease diagnosis. It typically includes 14 attributes including

- age
- sex
- chest pain type
- resting blood pressure
- cholesterol levels (*chol*)
- fasting blood sugar
- electrocardiographic results (*restecg*)
- maximum heart rate achieved
- exercise-induced angina
- ST depression (*oldpeak*)

The target value shows whether CVD is present or not. Second dataset focuses on CVD and contains clinical characteristics related to risk factors and health indicators. It typically includes attributes such as

- age
- gender
- height
- weight
- blood pressure (systolic (*APhigh*) and diastolic (*APlow*))
- cholesterol
- glucose
- smoking status
- alcohol consumption
- physical activity

Both datasets can be utilized to develop predictive models using machine learning algorithms. The common goal is to analyze the relationships between various attributes and the target variable (presence or absence of heart disease or cardiovascular disease and effective analysis) and build models that can predict disease outcomes for new individuals based on their characteristics.

B. Data Cleaning and Feature Extraction

Data cleaning and preprocessing are necessary to ensure data quality and consistency. This include managing outliers, handling missing values, standardising or normalising numerical characteristics, and, if necessary, encoding categorical variables.

Finding the appropriate features that offer more accuracy was done using a correlation matrix. We took into account the correlation between the attributes and the target attribute in the second [13] dataset. The attributes with positive and higher correlation were taken into account as the main features when developing the model. *age*, *APhigh*, *APlow*, *cholesterol*, and *glucose* were selected for the prediction model. However all 13 attributes were retained in 'Heart Disease Dataset' to predict the presence of the disease. While the analysis of disease is considered, dataset that has a true value for presence of disease were taken and 3 attributes, namely *restecg*, *oldpeak*, *chol* were selected.

C. Data Split

There were training and testing sets within the dataset. The testing set was used to assess the machine learning model's performance on unobserved data, whereas the training set was used to train the model. The typical split is around 80 percent for training and 20 percent for testing.

D. Model Development

It is necessary to choose a suitable machine learning algorithm. The type of problem and the data at hand determine whether an algorithm is appropriate. The various algorithm utilized for the detection of cardiovascular disease are as follows.

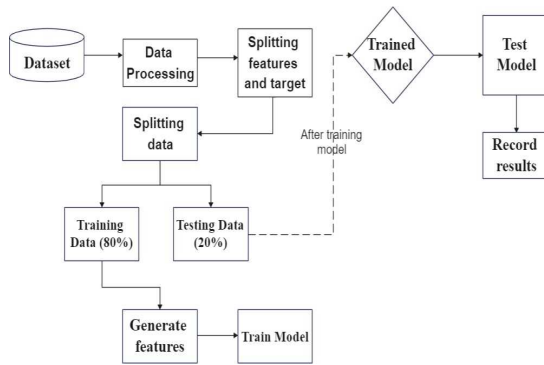


Fig. 1. Proposed methodology

Random Forest (RF): Several decision trees are combined in the RandomForestClassifier() ensemble learning technique to generate forecasts. A random subset of the training features and data is used to train each decision tree. The sum of the predictions from each individual tree yields the final forecast. The capacity of random forests to handle high-dimensional data and missing values is well known, and handle a mixture of numerical and categorical features. They can also provide feature importance measures, making them useful for understanding the importance of different features in the prediction.

K-Nearest Neighbors (KNN): A straightforward and efficient non-parametric classification algorithm is the KNN Classifier. It bases its predictions on the feature space's k-nearest neighbours' majority vote. KNN can handle binary and multi-class classification problems and is simple to comprehend and apply. Without the right adjustments, it performs poorly on imbalanced datasets and can be computationally expensive for large datasets.

Support Vector Machines (SVM): SVM is a algorithm that can be applied to tasks involving regression and classification. The goal is to identify the best hyperplane with the largest margin that divides data points belonging to different classes. By using kernel functions, it can handle both linearly and non-linearly separable data.

Decision Tree: It divides the data according to feature values and builds a binary tree from there. Decision trees can handle both numerical and categorical features and are simple

to understand and visualise. Without the proper pruning or regularisation techniques, they may not generalise well to new data and may be prone to overfitting.

Naive Bayes: It calculates the posterior probability of each class given the input features and makes predictions based on the highest probability. Naive Bayes is computationally efficient, particularly for high-dimensional data. It works well with categorical features and can handle continuous features by assuming a Gaussian distribution. However, it assumes independence between features, which may not hold in some cases.

Unsupervised algorithms are used for the evaluation of cardiovascular disease after heart disease prediction. They are listed below:

K-means Clustering:

The algorithm for unsupervised clustering divides data into K clusters according to how similar their feature values are. The algorithm assigns each data point iteratively to the closest centroid after randomly initialising the centroids of K clusters. The centroids are then updated based on the average of the assigned points. This process keeps going until it converges. K-means is a popular algorithm because of how easy and effective it is. It works well when the clusters are well-separated and have a roughly equal number of instances. However, it is sensitive to the initial centroid positions and may converge to local optima.

Gaussian Mixture Models (GMM):

GMM is a probabilistic model that represents the data as a mixture of Gaussian distributions, where each distribution represents a cluster. GMM assumes that the data points are generated from a combination of several Gaussian distributions, each associated with a specific cluster. It uses the Expectation-Maximization (EM) algorithm to estimate the parameters of the Gaussian distributions and assigns each data point to the most probable cluster. GMM is flexible and can model complex data distributions. It allows for soft assignments, meaning a data point can belong to multiple clusters with varying probabilities. However, it is computationally more expensive than K-means.

Hierarchical Clustering:

Through a series of iterative mergers and splits based on cluster similarity, hierarchical clustering is a technique that builds a hierarchy of clusters. Agglomerative (bottom-up) and divisive (top-down) hierarchical clustering are the two primary forms. While divisive clustering begins with all instances in one cluster, agglomerative clustering begins with each instance as a separate cluster and merges them based on their similarity.

E. Model Evaluation

Using assessment metrics relevant to the particular problem, evaluate the trained model's performance. Common metrics include accuracy using confusion matrix, k-fold cross validation and silhouette score incase of clustering. This stage aids in determining how well the model applies to fresh, untested data. A brief overview about the metrics used are as follows:

Confusion Matrix: This table displays the counts of various categories to provide an overview of a classification model's performance such as true positive (TP), true negative (TN), false positive (FP), and false negative (FN) predictions. The rows and columns of the confusion matrix represent actual and predicted class labels, respectively. The model's *accuracy*, *precision*, *recall*, and *F 1-score* can be evaluated using these matrix elements.

Accuracy: One common metric for assessing classification models is accuracy. It shows the percentage of cases that were accurately predicted out of all the instances. It is calculated as

$$A = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

and . However, it alone may not be sufficient in the presence of class imbalance, as high accuracy can still be achieved by predicting the majority class most of the time.

Silhouette Score: The silhouette score S is a metric used to assess the quality of clustering results. It measures how well, in relation to other clusters, each data point fits within the designated cluster. The score ranges for silhouettes from -1 to 1, greater values signifying improved clustering outcomes. For each data point, the silhouette score S is calculated as:

$$S = \frac{b - a}{\max(a, b)} \quad (2)$$

where a is the average distance to other instances within the same cluster and b is the average distance to instances in the nearest neighboring cluster. A high silhouette score indicates a well-clustered set of data points with large inter-cluster and small intra-cluster distances. A low silhouette score indicates that the data points may be incorrectly assigned to clusters or that the clustering may be inappropriate.

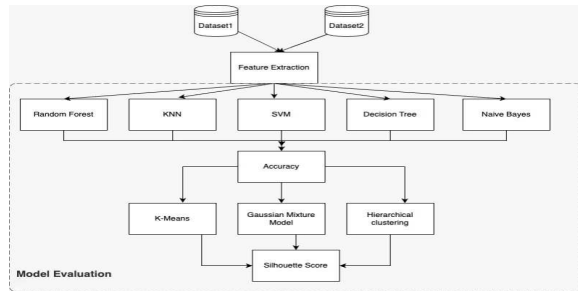


Fig. 2. Evaluation of machine learning models

F. Implementation

The dataset used includes instances of presence and absence of CVD. The work is implemented using "Google Colab" which is a web-based IDE for Python.

Five machine learning algorithms, namely, Random Forest, KNN, SVM, Decision Tree, Naive Bayes, were used to detect the presence or absence of CVD. The evaluation is done as shown in Fig. 2. Once the presence of the disease has been established, we assess the type of disease based on the three attributes— *restecg*, *oldpeak*, and *chol*. These attributes are

closely associated with determining the type of disease. We cluster the data into four groups, where each group determines the type of disease. *STABLE*, *ANGINA*, *STEMI*, *NSTEMI*, are four classes that have to be identified. K-Means provides the highest value of the silhouette score S while compared to the other algorithms.

IV. RESULTS AND DISCUSSION

The second dataset [13] upon selecting the attributes which had positive correlation with the target attribute has accuracy as shown in Table II when trained with various models. The input size post the preprocessing of data is 68,783 out of which 80 percent of data is considered as training data and 20 percent is test data. The TP, FP, FN, TN values for each of the algorithm are recorded in Table I

TABLE I
TP,FP,FN,TN VALUES FOR DATASET 2

SLNO	Algorithm	TP	FP	FN	TN
1	Random Forest	5360	1564	2247	4586
2	K-Nearest neighbors	5053	1871	2220	4613
3	Naive Bayes	5749	1175	2741	4092
4	Decision Tree	5408	1516	2388	4445
5	SVM	5796	1128	2633	4200

TABLE II
PREDICTION ACCURACY FOR DATASET 2

SLNO	Algorithm	Accuracy(In percentage)
1	Random Forest	72.51
2	K-Nearest neighbors	70.26
3	Naive Bayes	71.53
4	Decision Tree	71.65
5	SVM	72.66

When the dataset 1 [12] is taken into consideration with input size of 1026 out of which 80 percent is considered as training data and 20 percent is test data has better accuracy is obtained. The TP, FP, FN, TN and accuracy are as follows.

TABLE III
FOR DATASET 1

SLNO	Algorithm	TP	FP	FN	TN
1	Random Forest	102	0	3	100
2	K-Nearest neighbors	74	28	27	76
3	Naive Bayes	72	30	11	92
4	Decision Tree	102	0	3	100
5	SVM	62	40	25	78

TABLE IV
PREDICTION ACCURACY FOR DATASET 1

SLNO	Algorithm	Accuracy(In percentage)
1	Random Forest	98.53
2	K-Nearest neighbors	73.17
3	Naive Bayes	80
4	Decision Tree	98.53
5	SVM	68.29

Here we encounter similar accuracy values among Random forest and Decision tree. From Tables III and, IV, we can infer that the random forests are a good choice for CVD prediction. We go ahead and perform K-Fold cross validation up to 3 iterations to find a precise algorithm to predict the presence of disease. The observations of K-Fold cross-validation are summarized in the Table V.

TABLE V
K-FOLD CROSS VALIDATION

SLNO	Algorithm	Iteration 1	Iteration 2	Iteration 3
1	Random Forest	99.12	97.95	97.06
2	Decision Tree	98.2	97.95	96.18

Here it is clearly evident that Random Forest is a better fit. Additionally, compared to a single Decision Tree, Random Forest has a tendency to be more accurate. It aggregates many trees' predictions, which lowers bias and variance and improves performance all around.

The stage after the detection of presence of disease is to identify the disease. Here, various clustering algorithms are used to indicate the type of CVD a person is likely to suffer from. The silhouette score is used to come to a conclusion about the algorithm that is best suited. The values of the silhouette score are recorded in the Table VI shown below.

TABLE VI
SILHOUTTE SCORES OF VARIOUS ALGORITHMS

SLNO	Algorithm	Silhouette Score
1	K-Means	0.55
2	Gaussian Mixture Model	-0.20
3	Hierarchical Clustering	0.499

The type of disease is predicted using the K-Means algorithm because it has the highest silhouette score. K-means can handle big datasets and is computationally efficient. It is appropriate for applications with a lot of data points because of its linear time complexity.

V. CONCLUSION AND FUTURE WORK

As technology plays a bigger role in every aspect of our lives, we need to consider the potential applications of it in the medical industry. By using this application, doctors and their patients will be able to confirm this diagnosis and double-check their results. Additionally, this will aid medical students in their understanding of these disorders and their studies of diagnosis. The application's simplicity and use will encourage broader use and integration of Artificial Intelligence in the discipline of medicine. The method will be improved in a number of crucial areas in the next few years. First, a more thorough diagnostic tool would be made available by extending the illness classification beyond *STABLE ANGINA*, *STEMI*, and *NSTEMI*. Additionally, the system's performance can be enhanced by utilising cutting-edge artificial intelligence techniques like deep learning and reinforcement learning. Instant diagnostic results would be made possible by real-time analysis

capabilities, and information exchange would be streamlined by connectivity with electronic health records. The system's reach may be increased to create mobile and tele-medicine applications. The improvements in the ongoing model through frequent updates and retraining would guarantee its accuracy and dependability over time. These upcoming improvements will help the system continue to advance and become more precise and efficient at diagnosing problems.

REFERENCES

- [1] S.-Y. Cho, S.-H. Kim, S.-H. Kang, K. J. Lee, D. Choi, S. Kang, S. J. Park, T. Kim, C.-H. Yoon, T.-J. Youn *et al.*, "Pre-existing and machine learning-based models for cardiovascular risk prediction," *Scientific reports*, vol. 11, no. 1, p. 8886, 2021.
- [2] Cardiovascular diseases. <https://www.who.int/health-topics/cardiovascular-diseases>. Accessed: 23-06-26.
- [3] R. Katarya and P. Srinivas, "Predicting heart disease at early stages using machine learning: a survey," in *2020 International Conference on Electronics and Sustainable Communication Systems (ICESC)*. IEEE, 2020, pp. 302–305.
- [4] M. Limbitote, K. Damkondwar, D. Mahajan, and P. Patil, "A survey on prediction techniques of heart disease using machine learning," *International Journal of Engineering Research & Technology (IJERT)*, vol. 9, no. 6, pp. 2278–0181, 2020.
- [5] R. Bharti, A. Khamparia, M. Shabaz, G. Dhiman, S. Pande, and P. Singh, "Prediction of heart disease using a combination of machine learning and deep learning," *Computational intelligence and neuroscience*, vol. 2021, 2021.
- [6] S. Palaniappan and R. Awang, "Intelligent heart disease prediction system using data mining techniques," in *2008 IEEE/ACS international conference on computer systems and applications*. IEEE, 2008, pp. 108–115.
- [7] S. K. J. and G. S., "Prediction of heart disease using machine learning algorithms," in *2019 1st International Conference on Innovations in Information and Communication Technology (ICIICT)*, 2019, pp. 1–5.
- [8] N. M. Lutimath, C. Chethan, and B. S. Pol, "Prediction of heart disease using machine learning," *International journal Of Recent Technology and Engineering*, vol. 8, no. 2, pp. 474–477, 2019.
- [9] Y. K. Singh, N. Sinha, and S. K. Singh, "Heart disease prediction system using random forest," in *Advances in Computing and Data Sciences: First International Conference, ICACDS 2016, Ghaziabad, India, November 11-12, 2016, Revised Selected Papers 1*. Springer, 2017, pp. 613–623.
- [10] H. El Hamdaoui, S. Boujraf, N. El Houda Chaoui, B. Alami, and M. Maaroufi, "Improving heart disease prediction using random forest and adaboost algorithms," *International Journal of Online & Biomedical Engineering*, vol. 17, no. 11, 2021.
- [11] L. Yang, H. Wu, X. Jin, P. Zheng, S. Hu, X. Xu, W. Yu, and J. Yan, "Study of cardiovascular disease prediction model based on random forest in eastern china," *Scientific reports*, vol. 10, no. 1, p. 5245, 2020.
- [12] Cardiovascular diseases. <https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>. Accessed: 23-06-26.
- [13] Cardiovascular diseases. <https://www.kaggle.com/datasets/aiaiaidavid/cardio-data-dv13032020>. Accessed: 23-06-26.